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This is the author's manuscript

Original Citation:

Availability:

This version is available http://hdl.handle.net/2318/2113653 since 2026-01-12T15:20:32Z

Published version:

DOI:10.1080/10447318.2025.2491009

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19 March 2026

How do People Develop Folk Theories of Generative AI Text-to-image Models? A Qualitative Study on How People Strive to Explain and Make Sense of GenAI

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Generative Artificial Intelligence (GenAI) text-to-image models have made significant progress in emulating human-like outputs. However, understanding the inner functioning of these models remains a challenge due to their complexity and black-box nature. It has been observed that individuals naturally develop informal conceptualizations, termed "folk theories", to explain the behaviors of algorithmic systems. The specific nature of GenAI text-to-image models, which are obscure in their working principles, yet carry out activities that are peculiar to humans, makes it interesting to investigate people's theorization about this technology. With this aim, we conducted a qualitative interview study with 20 participants and observed how they accounted for the outputs of Stable Diffusion. The study findings show that participants developed a wide spectrum of conceptualizations, including folk theories that appear distinctive of GenAI text-to-image technology, also ascribing to the model a variety of "mental states". Furthermore, we found that theory building follows different inductive and deductive trajectories, with participants employing diverse strategies to explain the functioning of the technology.

Keywords: AI, Stable Diffusion, anthropomorphizing, folk theories, Generative AI, XAI, Explainable AI

Declarations of interest: none

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This is the accepted version of the article published by Taylor & Francis and available at: https://doi.org/10.1080/10447318.2025.2491009

To cite this article: Chiara Di Lodovico, Federico Torrielli, Luigi Di Caro & Amon Rapp (2025) How Do People Develop Folk Theories of Generative AI Text-to-Image Models? A Qualitative Study on How People Strive to Explain and Make Sense of GenAI, International Journal of Human–Computer Interaction, 41:23, 14846-14870, DOI: 10.1080/10447318.2025.2491009

1 INTRODUCTION

The field of Generative Artificial Intelligence (GenAI) has recently witnessed remarkable progresses in the development of models that can emulate the creation of human-like outputs, including images, text, speech, and music (Muller et al., 2022; Bandi et al., 2023).

In the context of GenAI text-to-image models, technology is able to produce high-quality synthetic images from natural language textual inputs, also called "prompts", using deep neural networks trained on huge multimodal datasets (Vimpari et al., 2023; Oppenlaender, 2022; Tang et al., 2023). The introduction of models adopting easy-to-use interfaces, like DALL·E 2[1], Midjourney[2], and Stable Diffusion[3], has significantly reduced the barriers to the adoption of this technology, by enabling anyone, regardless of their technical knowledge, to swiftly produce digital images and artworks based on simple text input (Oppenlaender, 2022; Wang et al., 2023; Liu & Chilton, 2022). Since their appearance, these models have been used by millions of registered users, who have produced billions of images (OpenAI, 2022; Stability AI, 2023; Codewatchers, 2023). Their popularity attests that the impact of these new technologies may reach a broader population who may be continuously exposed to AI-generated images.

Similarly to other GenAI technologies, text-to-image models are known to produce outputs that are often difficult to decipher in terms of how and why they were produced by the machine, even for people with a strong technical background (Ooi et al., 2023; Capel and Brereton, 2023). For this reason, these models are frequently characterized as "black boxes" because people can observe the inputs and outputs of the technology but may not be able to figure out the mapping between them (Chromik et al., 2021).

In response to this challenge, a growing body of research in the field of explainable AI (XAI) focuses on the development of strategies and techniques that enhance the transparency and comprehensibility of AI systems' behavior (Rozenblit & Keil, 2002; Chromik et al., 2021). However, people may produce explanations of their behavior despite their obscure functioning. In fact, individuals may spontaneously produce a variety of "folk theories" of technology. These theories are informal conceptualizations addressed to explain the outcomes, working logic, effects, or consequences of algorithmic systems (Rader and Gray 2015, DeVito et al. 2017), and can be *"fleeting, inconsistent, only partly acknowledged or articulated, based on limited or*

[1] https://openai.com/dall-e-2/
[2] https://www.midjourney.com/home/
[3] https://stablediffusionweb.com/

misleading knowledge" (Ytre-Arne & Moe, 2020, p. 811). In any case, folk theories have a profound impact on how users perceive, adapt, and interact with technological systems, also affecting their acceptance or refusal (e.g., DeVito et al., 2018).

It comes as no surprise, therefore, that the Human-Computer Interaction (HCI) community has extensively devoted attention to how individuals produce folk theories of technology. In this context, researchers mostly focused on the functioning of algorithms underlying the work of social networks and social media (Eslami et al. 2016, French & Hancock 2017, DeVito et al. 2017, DeVito et al. 2018, Karizat et al).

However, these technologies consistently differ from black-box technologies like GenAIs, being much opaquer in their inner working principles and, more importantly, producing outputs that are similar to those created by humans. For this, it may be reasonable to hypothesize that the theories developed by people to explain the functioning of GenAI may differ from those addressed by previous research, showing peculiarities in both the kind of theories produced and in how they are produced. In this sense, it becomes fundamental to study folk theories in the context of GenAI: in particular, studying the field of text-to-image models may yield important insights given the increasing impact that these technologies have (and will soon have) on our everyday life, and the role that folk theories may play in shaping the users' understandings and perceptions of the technology.

To this aim, we explored how people with diverse backgrounds and with a different degree of previous knowledge and experience of AI and GenAI conceptualize a text-to-image-model and its "ability" to generate images from text. In doing so, we sought to uncover not only the kind of folk theories that they developed regarding the model's functioning, but also *how these theories were formed*, trying to answer the following research questions:

(RQ1) What kind of folk theories do people develop about a GenAI text-to-image model starting from its outputs?

(RQ2) How are these theories formed, and how does the folk theory development process unfold over time?

To answer these questions, we conducted a qualitative study using semi-structured interviews with 20 participants. Instead of making them directly interact with a GenAI text-to-image model, which could have produced a variety of human-GenAI interactions and generated images, we opted for providing participants with a series of predefined stimuli: each participant was presented with 20 different images produced by Stable

Diffusion and the corresponding term used to generate them and was asked to share their perceptions and understandings about the technology and its creations. Even though this choice may appear "artificial" at first sight, because participants did not develop their folk theories on the basis of actual interaction with the model, it reflects how people produce explanations about a variety of world phenomena: in fact, people commonly develop theories of phenomena that they do not directly experience (e.g., they "encounter" the phenomenon only in a "mediated" way, like a volcanic eruption that they may see in a movie); likewise, they may produce folk theories about technologies with which they did not interact (Chromik et al., 2021), or which do not even exist (Grill & Andalibi, 2022). By providing a series of standardized stimuli, we homogenized "the material" on which participants could develop their folk theories, which allowed for the exploration, for example, of the role that individual differences may play in generating them and whether the "same outputs" could lead to different understandings. In so doing, we also provided participants with a "heatmap" for each presented image, namely, a representation that shows what elements of the image were more important for the GenAI in generating the image starting from specific terms. These representations are considered the state of the art in XAI methods for text-to-image models and are meant to increase the transparency of the functioning of the GenAI model: by offering these "aids" to the participants, we aimed to investigate whether current XAI tools could help individuals develop their understanding of the model.

Study findings show that participants produced a variety of folk theories, which can be differentiated between *mechanistic theories*, addressed to explain the working principles and parts of the text-to-image model, and *essentialist theories*, seeing the model as an entity endowed with intentionality and agency. Moreover, we discovered that folk theory development unfolds progressively and through trial and error. Participants followed different paths to build their own folk theories, pursuing either a linear additive process, where each step of theory development is grounded on the previous one, or a circular iterative process, in which phases of theory building are recurrently alternate with phases of assessment and theory refinement or substitution. In so doing, they relied on a variety of strategies to explain the model's behavior, including the reference to more familiar technologies, the identification with the model, and the connection with their own practices and experiences, as well as with the abilities that they supposed to be peculiar to humans.

These findings highlight three main contributions. First, we described a variety of folk theories that people develop in relation to GenAI text-to-image models, which in part differ from those highlighted by previous

research (DeVito et al., 2017; DeVito, 2021; Grill & Andalibi, 2022; Karizat et al., 2021). In particular, the theories of GenAI as human-like, GenAI as a human byproduct, and GenAI as other-than-human appear to be specific to text-to-image models, attributing "mental states" to the machine, like intentionality and agency, and taking the human being as a point of reference to understand its "essence" and abilities.

Second, we offered a nuanced perspective on how people form folk theories about technology, complementing and extending previous research on this line (DeVito et al., 2018; DeVito, 2021). In this sense, we highlighted the extreme malleability of both the theory formation process and the theories produced, which are multiple, changeable, and amendable, in a much greater degree than that has been observed in relation to social media (DeVito et al., 2018; DeVito, 2021).

Third, we pointed out that participants with different backgrounds may differently account for the creations of the text-to-image model, relying on diverse strategies and showing different attitudes. In particular, participants with technical skills were more cautious in formulating hypotheses on the inner functioning of the model and mostly referred to other AI systems to explain its behavior; while those without a technical background were more confident in formulating mechanistic theories, connecting with other non-AI technologies, like search engines, and generating more naïve explanations. All the participants, however, engaged in producing theories that see the model as an intentional entity.

2 BACKGROUND

2.1 Generative AI Text-to-image Models

Generative AI text-to-image models, like DALL·E 2[4], Midjourney[5], and Stable Diffusion[6], use deep neural networks trained on huge multimodal datasets to produce high-quality digital images from natural language textual inputs, also called "prompts" (Tang et al., 2023; Zhang et al., 2023b). In particular, the introduction of Stable Diffusion in 2022 (Rombach et al., 2022) led to the creation of accurate and high-resolution images, establishing the state-of-the-art in text-to-image GenAI (Vimpari et al., 2023). Shortly thereafter, other state-

[4] https://openai.com/dall-e-2/
[5] https://www.midjourney.com/home/
[6] https://stablediffusionweb.com/

of-the-art systems such as Midjourney and DALL·E 2 were subsequently released, increasing the popularity of these models (Vimpari et al., 2023).

Despite the overly complex inner functioning of this technology, the introduction of easy-to-use interfaces has allowed anyone to quickly produce unique digital images and artworks, regardless of their design skills or technical knowledge (Wang et al., 2023; Liu & Chilton, 2022). Notably, by September 2022, DALL·E 2 had 1.5 million active users (OpenAI, 2022), while Stable Diffusion had more than 10 million users (Stability AI, 2023); by October 2023, Midjourney had an average of 1.5 million daily active users (Codewatchers, 2023). These numbers attest that these technologies could produce an enormous social impact: GenAI images find now widespread use on social networks (Oppenlaender, 2022), achieve recognition in art competitions (Roose, 2022), and feature on magazine covers (Liu, 2022).

Given the social significance of this phenomenon, there is a notable increase in interest in text-to-image GenAI within the HCI research community. Several recent studies explore the adoption and utilization of these models among various user groups, including online communities (Buraga, 2022), creative professionals (Inie et al., 2023; Vimpara et al., 2023; Ko et al., 2023; Zhang et al., 2023a), and laypeople (Wang et al., 2023), as well as people's perceptions of the generated images, also comparing AI-generated and human artworks (Hong & Curran, 2019; Ragot et al., 2020; Ting et al., 2021; Gu & Li, 2022).

A recurrent theme across this body of research is the opacity of GenAI models, offering limited interpretability of their inner working compared to traditional programming systems. As highlighted by Zamfirescu-Pereira et al. (2023), these technologies bring a recurring challenge for users and experts, which may be summarized in the question: *"But why did it do that?"*. The question points at a significant concern, as the opacity of these models may hide potential biases, also questioning their effective and fair use: in fact, it is difficult to trust, "collaborate with", and take decisions on the basis of outputs produced by technologies that cannot be understood in their working principles (Bird et al., 2023). Within the growing community of scholars investigating GenAI models, there is, therefore, an increasing emphasis on the necessity to find methodologies capable of shedding light on their intricate "decision-making" processes.

2.2 Text-to-image Generative AI and explainability

GenAI text-to-image models are known to produce outputs that are complex in terms of comprehension and explanation of how they were produced (Ooi et al., 2023). Even when trained with publicly accessible

resources, it can be extremely challenging to decipher their acquired knowledge and underlying behavior (Capel & Brereton, 2023; Lee et al., 2023). In fact, even in the case of open-source models like Stable Diffusion, where the overarching system architecture and high-level operational principles are disclosed, a significant gap remains in comprehending the inner functioning of the system, particularly how it transforms input A (such as the text prompt "a dog with a green hat") into output B (the image of "a dog with a green hat") (Chromik et al., 2021).

The growing need to address their black-box nature has encouraged the development of the eXplainable AI (XAI) research field (Ehsan & Riedl, 2020). XAI is dedicated to making the inner workings of AI technologies interpretable and explainable, by developing an array of tools, methodologies, and frameworks aimed at facilitating human understanding of the decisions and predictions generated by AIs (Speith, 2022; Chromik et al., 2021; Capel & Brereton, 2023).

In the context of GenAI text-to-image models, researchers approached XAI matters from different perspectives. Some approaches explore how the features of textual inputs may influence the output by leveraging prompt engineering, attempting in this way to unveil aspects of the hidden behavior of the models (Oppelaender, 2022; Liu & Chilton, 2022; Wang et al., 2023). Other studies, instead, attempt to support non-expert users in understanding how GenAI text-to-image models work by developing global explanation that make their overall inner logic more accessible (e.g., Lee et al., 2023).

On this line, further research focuses on attention visualization methods, a type of explanation that provides cues into the reasons why a specific prediction or decision of the model took place (Mansimov et al., 2016; Sun et al., 2022). The most popular attention visualization methods for visual explanation are heatmaps (Zeiler and Fergus, 2014). A heatmap typically appears as a gradient-based visual overlay on an image the intensity of each pixel's color conveys the degree of importance that the pixel had in representing the input elements. Alqaraawi et al. (2020) and Kim et al. (2023) conducted user tests on heatmaps and saliency maps to evaluate their effectiveness in explaining AI-based model behavior. The results of both studies indicate that these methods elicited contrasting responses, with some participants finding heatmaps intuitive and easy to understand and others considering them un-intuitive and uninformative.

By and large, XAI studies primarily aim to utilize XAI methods to allow for an optimal utilization of GenAI models. However, as noted by Petsiuk et al. (2018), an AI system could behave differently from a human and

learn to use cues from the background (e.g., using grass to detect cows) or other cues that are non-intuitive to humans. In this context, instead of looking for ways to "open the black box", we should be more interested in opening "the black box of one's own assumptions about knowing algorithms" (Bucher 2018, p. 46) investigating individuals' theories about the working logic of GenAI systems.

2.3 Folk Theories

Causal thinking plays a crucial role in human cognition and learning. From early childhood, humans exhibit a natural inclination to understand, predict, and explain the reasons behind events and phenomena – essentially, to understand "how and why things happen" (Gelman & Legare, 2011, p. 4). To navigate the complexities of the everyday world and engage with intricate systems, ranging from interactions with other humans to natural phenomena and technologies, individuals develop conceptualizations that influence their beliefs, behaviors, and decisions (French & Hancock, 2017).

In this scenario, GenAI text-to-image models, which are intentionally designed to produce outputs that closely resemble human creations, potentially intensify the human inclination to develop theories about their operational principles, akin to our conceptualizations of other humans' behavior (Bucher, 2018). As previous research demonstrates, individuals may anthropomorphize, attribute human meanings to algorithmic systems (Jones et al., 2023), ascribing humanness and intelligence to AI agents (Rapp et al., 2023b; Rapp et al., 2021), despite acknowledging their artificial nature (Bender & Korrel, 2020). For this, there is an increasing interest in research aimed at comprehending individuals' beliefs, perceptions, and conceptualizations of algorithmic systems. This research stream delves into various matters, including the examination of people's mental models of AIs (Kelly et al., 2023), the exploration of "algorithmic imaginaries" (Bucher, 2018), and the investigation of "folk theories" (Rader & Gray, 2015) about AI-powered technologies.

Folk theories are particularly important because they represent the main users' strategies for understanding complex technological systems: they often act as a high-level frame for shaping their expectations, thus being a useful window into understanding acceptance and rejection of technologies, unveiling people's underlying concerns or preconceptions towards them (DeVito et al., 2017). Also referred as "intuitive theories" (Gelman & Legare, 2011), folk theories are not-formal explanatory causal theories that individuals develop to make sense of the inner workings, consequences, and impacts of technological systems (Rader & Gray, 2015; DeVito

et al., 2017), contributing to shaping how individuals perceive, understand, and interact with them (Jones et al., 2023; DeVito et al., 2018).

People tend to formulate folk theories in a manner similar to how they develop theories about the causes of events in the world. In this sense, as people may produce theories about phenomena even without directly experiencing them, like collisions of asteroids that may be "encountered" only in a mediated way (e.g., being seen in a documentary or in a movie), they may also develop folk theories about technology even without directly interacting with it (e.g., a particle accelerator) (Chromik et al., 2021) and even about technology that does not presently exist (Grill & Andalibi, 2022).

Previous studies pointed out that folk theories may be re-evaluated as new information is gained and further interactions with the system are made (Gelman & Legare, 2011), resulting from the complex aggregation of multiple and disparate sources of information (DeVito et al., 2018). Folk theories may also have diverse levels of "malleability" depending on the individual's level of confidence in their understanding and can be updated when "new, and even contradictory, information" is available (DeVito et al., 2018). In this scenario, inconsistencies (Gelman & Legare, 2011) and expectation violation (Rader & Gray, 2015; Eslami et al., 2015) may motivate and stimulate folk theory construction to reconcile a problem to be solved (Gelman and Legare, 2011).

Investigating users' folk theories, therefore, may provide valuable insights into their conceptualization of systems, which may subsequently impact on how they use, perceive, and interact with them. In this context, much research has been devoted to identifying specific folk theories that users produce to explain the functioning of certain technologies. For instance, French and Hancock (2017) found four primary folk theories that users apply to Facebook News Feed and Twitter, connecting them to the positive (the Rational Assistant and the Transparent Platform) and the negative feelings (the Unwanted Observer and the Corporate Black Box) that they experience towards them. In parallel, DeVito et al. (2017) and DeVito (2021) highlighted that there exist abstract theories about the overall algorithmic system (i.e., Twitter), which rely on a general sense that an algorithm is something that can cause something to happen, as well as operational theories, which attempt to theorize how an algorithm might actually operate. Similarly, Siles et al. (2020) explored two contrasting folk theories concerning algorithmic music recommendations on Spotify. One portrayed Spotify as a "social being providing recommendations through surveillance", while the other envisioned it as a "computational

machine offering an individualized musical experience through appropriate training". In a similar vein, Grill and Andalibi (2022) explored conceptions and expectations of hypothetical emotion recognition technologies used on social media, asking participants to envision themselves in speculative scenarios where this technology is a reality: participants made sense of emotion recognition based on popular myths about big data and AI.

By contrast, other research focuses more on how folk theories are formed. For example, DeVito (2021) studied how non-expert users adapt their theorizations to changes in algorithmically-driven social platforms. She found that users re-evaluate their folk theories in response to platform change and that both the complexity of the user's folk theorization and their overall relationship with the platform impact the theorization process. Likewise, DeVito et al. (2018) highlighted that people may generate folk theories from multiple, often disparate, pieces of information from different sources, comparing these multiple sources of information with each other to verify accuracy. In this sense, the sense-making process may consist of constantly modifying one's understanding to integrate new information, whereby the formed folk theories may be more or less rigid.

However, most of the studies exploring folk theories relate to web services, social media platforms and social feeds (e.g. French & Hancock, 2017; DeVito et al., 2018; Siles et al. 2020). These technologies consistently differ from GenAI models, which are much opaquer in their underlying functioning and produce outputs that are similar to those created by humans: in fact, while social media algorithms do a "kind of work" that may not be associated to human activity (i.e., delivering media that can be appreciated by the user), GenAI technologies are now performing tasks that until recently were thought to pertain exclusively to humans with results that approach, and sometimes are indistinguishable from, human productions. Previous research highlighted that many people already perceive a degree of consciousness in GPT-3 (Scott et al., 2023). In this sense, the kind of theories produced by people to explain the inner functioning of GenAI text-to-image models may also differ from those utilized to understand the behavior of other systems, thus appearing to be worthy of being further explored. This is even more important if we consider the increasing impact that these technologies have (and will soon have) on our everyday life, as well as the role that folk theories may play in shaping the users' understanding and perception of technology. Shelby et al. (2024) explored how artists conceptualize text-to-image models, identifying folk theories of use, harm, and harm reduction: theories of use concern the divergent ways models are employed in creative practice as a creative medium or a mundane tool; theories of harm concern models either as something harmed by engineering and product policy efforts to

eliminate failures and limit functionality, or as a source of harm; theories of harm reduction focus on protecting the models, by increasing transparency around model development and artist control over them. Although this work provides interesting insights, it focuses on folk theorizations that are deeply entangled with artists' creative practices, without capturing how the broader population may conceptualize this technology.

In this study, we precisely investigate the folk theories that individuals with different backgrounds and degrees of direct experience with GenAI develop. Instead of requiring participants to interact with a text-to-image model, we presented them with the model's outputs, namely a series of GenAI images and their corresponding heatmaps. Inspired by Siles et al. (2020), we have recruited participants with diverse levels of technical expertise and professional backgrounds. Moreover, aligning with DeVito et al.'s (2017) work, we included participants who are newly or vaguely acquainted with AI technology and may have not already engaged in theorizing about how algorithms operate. Differently from previous research, we observed participants in the "ongoing theorization process": instead of asking them to recount how they explained and interpreted their past interactions with a system, we observed how people made sense of different outputs of the technology as they were presented to them. In this way, we aimed to investigate the entire theorization process, from the initial encounter with the technology's creations, to later reflections arising from the acknowledgment of different outputs.

3 METHOD

To investigate people's folk theories about GenAI text-to-image models we used semi-structured interviews providing the GenAI outputs as stimuli. Originally, the study had two main goals. First, we aimed to investigate how individuals may react to the images that a GenAI text-to-image model can produce, in terms of perceptions and emotions that they may elicit. Second, we wanted to study whether and how people formed folk theories about the functioning of this kind of opaque technologies. While the recounting of the findings relating to the first goal goes beyond the scope of this article and won't be reported here (Rapp et al., 2025b), in the following Sections we will focus on the second goal, describing the participants' attempts to produce explanations about GenAI text-to-image technology starting from its creations.

We recruited twenty participants with diverse backgrounds, inviting them to look at twenty different images generated by Stable Diffusion 1.5 starting from specific text prompts. Instead of making them directly interact with the GenAI text-to-image model, we thus opted for providing participants with a series of homogenized

stimuli: in this way, we standardized the "material" on the basis of which the participants could develop their own theories, which allowed us to explore, for example, whether the "same outputs" could lead to different participants' understandings. This way of investigating folk theories is also in line with research that studied how people form theories of technology that they did not directly experience, for example because it does not exist yet (Grill & Andalibi, 2022).

In line with previous research (Tang et al., 2023), we provided a variety of stimuli (i.e., the images), as comparing different GenAI outcomes might better enable participants to formulate inferences about the model's functioning, considering variations in its outputs and their unique characteristics. Moreover, we paired each image with its corresponding heatmap to explore whether and how these tools may support people in developing their understandings: heatmaps allow people to identify which parts of the input are more influential into the generated output, potentially helping them in discerning what aspects are important for the AI in creating the image (Kim et al., 2023). A heatmap is a gradient-based visual overlay on an image, in which the intensity of each pixel's color conveys the degree of importance that the pixel had in representing the input elements.

3.1 Materials

To generate the images presented during the study, we used Stable Diffusion for its open-source nature and its capacity to create both images and their corresponding heatmaps, by integrating the diffusion attentive attribution maps (DAAM) module, as introduced by Tang et al. (2023). We employed a consistent prompt formula across all the creations, employing the prompt "Representation of ____" (e.g., "Representation of anger"). This choice was made considering that using only one term (e.g., "Sea") would likely have produced details characterizing the subject (e.g., waves). By adding "Representation of ____," we encouraged the model to explore various ways of representing the subject. Moreover, we considered the Classifier-Free Guidance (CFG) Scale (Ho & Salimans, 2021), which makes the GenAI attribute varying levels of image's fidelity to the prompt (from 20, indicating extreme fidelity, to 1, where the GenAI generates more diverse images). We used a CFG value of 7.5, since it is the standard value used in text-to-image model research (Wang et al., 2023), granting a balance between fidelity to the prompt and a degree of "creativity" in the image generation process.

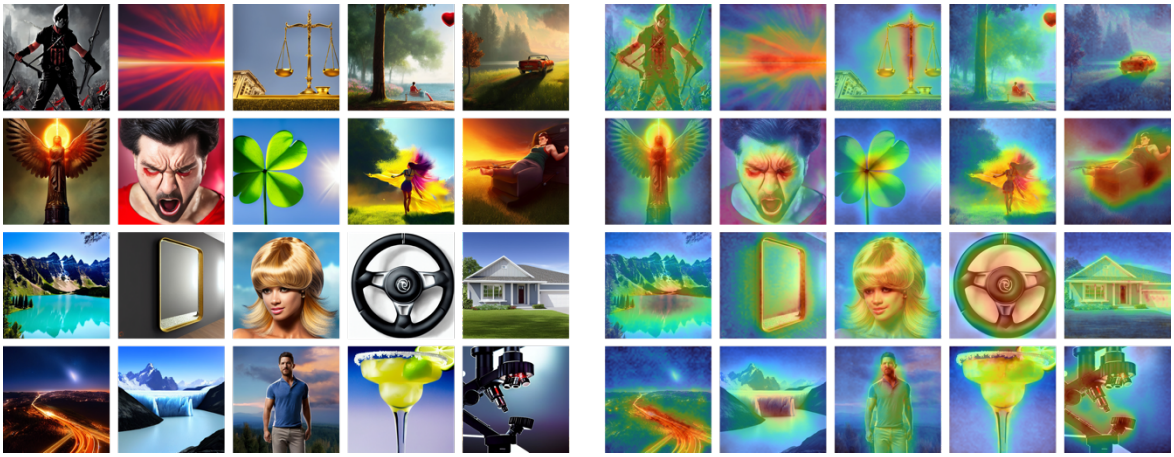

Fig. 1. Images and heatmaps generated via Stable Diffusion 1.5. From top left to bottom right: Anarchy, Infinity, Justice, Love, Nostalgia, Wisdom, Anger, Luck, Euphoria, Laziness, Lake, Mirror, Wig, Steering wheel, House, Night, Glacier, Man, Margarita, Microscope.

The choice of terms for creating the images employed a purposeful sampling method, which states that researchers actively select the most productive sample to answer their research questions, identifying the variables that might influence the study outcomes on the basis of their practical knowledge of the research area and previous literature (Marshall, 1996). We thus followed practices performed in prior research on text-to-image models (Liu & Chilton, 2022; Oppenlaender, 2022) and considered abstractness and concreteness criteria to diversify the sample, because abstract images are more open to interpretation than concrete images and thus could potentially elicit different responses. We used the concreteness rating database defined by Brysbaert et al. (2013), which includes 40,000 English words rated on a scale that ranges from abstract (1) to concrete (5) as a starting point for the sample selection. We first selected abstract terms with ratings ranging from 1 to 2.5 and concrete terms with ratings between 3.5 and 5. Then, we refined the selection by taking into account i) *the comprehensibility of the term*, preferring those terms that were likely universally understandable, since terms difficult to understand or that are not know by the general population could also be not known by several participants; ii) *the potential for visual representation*, so that the generated image could be connected to the original term. In practice, this selection process underwent multiple iterations. After extensive discussions among the four authors, we defined an initial pool of 200 terms. Then we engaged four external researchers asking them to report on their understandability and potential for visual representation. We

eventually chose those 20 terms that did not raise any concern or disagreement from any of the researchers (i.e., terms for which all researchers agreed were either concrete or abstract, as well as easily understandable and depictable), then creating the corresponding images and heatmaps (Fig. 1).

3.2 Participants

We followed a purposeful sampling method for selecting participants (Marshall, 1996), differentiating the sample mainly along the background/profession dimension and previous experience with GenAI or AI. In this way, we wanted to explore how people with different levels of technical understanding and use of text-to-image GenAI models generated explanations about the model's functioning. Participants ranged from AI experts with a technical background and designers with a creative background, to people with no knowledge of AI and no previous use of GenAI systems. To increase the sample heterogeneity, we further balanced it along gender. Participants' demographics are detailed in Table 1. All the participants were Italian. To settle the sample size to 20 participants, we followed a theoretical saturation principle (Bowen, 2008): we stopped recruiting participants when we realized that additional data would not have produced substantial new results for the objectives of the study. In any case, the sample size is aligned with common practices in qualitative research (Marshall et al. 2013) and other HCI qualitative studies with similar purposes (e.g., Sadek et al., 2023; Deng et al., 2023). Participants were recruited via email and snowball sampling.

Table 1. List of participants, background, gender, age and previous use of AI text-to-image systems

Code	Background	Gender	Age	Interview duration	Profession	Education	Previous use AI text-to-image	Previous use of other Gen-AI systems
P1	Tech	F	28	2h10	Research Fellow in Computer Science	Master's degree	None	ChatGPT
P2	Tech	M	26	2h	Research Fellow in Computer Science	Master's degree	None	None
P3	Tech	M	26	2h10	Master Student in Computer Science	Bachelor's Degree	None	ChatGPT
P4	Tech	F	30	2h	PhD student in Computer Science	Master's degree	None	ChatGPT
P5	Non-Tech, Non-Design	F	28	2h05	Data Governance Specialist	Bachelor's Degree	None	None
P6	Design & Tech	F	30	2h05	Graphic Designer	Master's degree	Midjourney, DALL·E 2	ChatGPT
P7	Design	F	34	2h10	Product Designer	Master's degree	None	None
P8	Non-Tech, Non-Design	M	69	2h	Cultural operator	Master's degree	None	None
P9	Non-Tech, Non-Design	F	63	1h40	High School Professor	PhD	None	ChatGPT
P10	Non-Tech, Non-Design	M	71	1h50	Psychologist	Master's degree	None	None

Code	Background	Gender	Age	Interview duration	Profession	Education	Previous use AI text-to-image	Previous use of other Gen-AI systems
P11	Non-Tech, Non-Design	M	34	2h30	Social worker	Bachelor's Degree	None	None
P12	Non-Tech, Non-Design	F	34	1h40	Psychiatrist	Master's degree	None	None
P13	Non-Tech, Non-Design	F	37	2h15	Doctor	PhD	None	None
P14	Design	F	35	2h10	Product Designer	Master's degree	None	None
P15	Non-Tech, Non-Design	M	28	1h45	Musician	Bachelor's Degree	Midjourney, DALL·E 2, Stable Diffusion	ChatGPT
P16	Design	F	33	2h40	Research Fellow in Design	Master's degree	ImaginAI	ChatGPT
P17	Design	F	32	2h	Product Designer	Master's degree	Midjourney, DALL·E 2	ChatGPT
P18	Design	M	32	2h30	Designer and PhD student in Design	Master's degree	Midjourney, DALL·E 2	ChatGPT
P19	Des & Tech	F	32	1h50	PhD student in Design	Master's degree	None	None
P20	Design	F	38	2h20	UX Designer	Master's degree	None	ChatGPT

3.3 Procedure

Firstly, the first author briefly described the study, only explaining that it had the aim of exploring their perception of certain images. In this introduction, we still did not want to reveal that the images were artificially generated.

Secondly, we presented participants with 20 images in a randomized order, one at a time, to minimize any possible presentation order effect. Each participant was invited to describe the subject of the representation depicted in the image using a single term (i.e., What does this image represent to you?), and to explain the reasons lying behind their answer (e.g., Why does this image represent love to you?). Furthermore, we asked them to describe the image and recount their reactions to it. We emphasized that there were no right or wrong answers, and we assured participants that the study was not designed as an intelligence test. This phase aimed to collect the reactions of the participants to the presented images before they were told that they were artificially produced by a GenAI. More precisely, we wanted to explore their "natural" way of interpreting the images, without them feeling compelled to guess the term used by the text-to-image model to generate the images. At the same time, we aimed to check whether their "interpretive process" diverged from that of the model (by mentioning a completely different term from the one used in the prompt) or converged with it (by mentioning the same term or a similar one). This process was repeated for all the 20 images produced by the GenAI.

Then, we told the participants that the images were generated by a GenAI from one specific term, which represented the subject of the image. We reviewed each image in the same order as before while disclosing

their original prompt. For each image, we gave the participants time to reflect on the similarities or differences between the term they mentioned in the previous phase of the study, which they believed represented the subject of the image, and the term used as the prompt, as well as on the reasons behind these similarities or differences. Then, we left participants free to share their thoughts about the text-to-image model and the image, adopting a sort of thinking-aloud protocol. In this way, we wanted to understand whether and how the participants formulated a "folk theory" about how the GenAI could produce the presented images, explaining its supposed functioning. By adopting a sort of thinking-aloud protocol, we tried not to force participants to engage in theory building; rather, we allowed them to freely express what they thought in the way they considered appropriate. For each image, we also showed the corresponding heatmap introduced with the following explanation: "*Heatmaps show what elements of the image are most salient/important for the GenAI to generate an image representing a given term. If you ask the AI to represent 'a dog', the heat map enables you to see which parts of the image are seemingly more relevant for the AI in generating the representation of that term*". We then invited participants to reflect on what such heatmaps meant to them and left them free to report on any further thought that they considered important. In this way, we aimed to investigate whether and how the heatmaps changed their initial interpretations about the model and the generation process.

Finally, we presented participants with all the images displayed on a table and conducted a semi-structured interview to explore their general feelings about and understandings of the images, the GenAI and the text-to-image generation process. We left the interview as open as possible to leave the participants free to tackle the themes that they considered most important in relation to the GenAI and its creations (e.g., By looking at all these images do you find some recurrent elements? In general, which ones have struck you the most, and why?). The study lasted about two hours. The participants were not compensated for their participation. The study was approved by the ethical board of the University of Turin (n. 0096227).

3.4 Data analysis

The participants' responses and interviews were recorded and transcribed verbatim. We then conducted a thematic analysis (Braun & Clarke, 2006), which is a widely used analytical method in HCI, because of its flexibility and independence of a specific theory and epistemology, using an inductive, rather than a deductive, stance (Patton, 1990). The analysis was conducted with the support of Quirkos, a software for qualitative analysis. The first author first read all the transcripts multiple times to get familiarity with the data.

Subsequently, initial codes were generated, by identifying data features that she considered relevant to capture the participants' understandings of the model. Data were broken down into sentences and short paragraphs and then labeled with the corresponding code. Then, the initial codes were grouped into emerging themes, going deeper into the latent content of the data to identify the participants' conceptualizations about the GenAI and its creations, as well as how they explained the GenAI's functioning and addressed the inexplicability of certain productions.

During the analysis, the first author engaged in an iterative discussion with the last author, who also read all the transcripts multiple times. They reviewed the codes and themes together discussing their formulation (e.g., whether a label clearly pointed at a certain concept) and application (e.g., whether the code well identifies a data pattern) (MacQueen et al. 1998; McDonald et al. 2019). As is common in qualitative research embracing an interpretative approach (e.g., Yardley 2000; Harry et al. 2005; Braun & Clarke 2013) as well as within HCI (e.g., Jun et al. 2018 ; Yang & Neustaedter 2018), we did not aim to reach a formal intercoder reliability, so no numerical reliability rating is reported: our goal was to reach consensus on a shared intersubjective interpretation, where codes and themes were debated, as well as their application, until the two authors agreed on their appropriate usage (Harry et al. 2005). The analysis led to the identification of 76 initial codes and 16 themes. These were further amalgamated into seven overarching themes, part of which (those related to the participants' folk theories) are recounted in the following Section. We will extensively report participants' quotations in line with standards of qualitative research, to make the reader grasp their understandings and perceptions from their own words.

4 RESULTS

We identified four overarching themes related to the development of folk theories during the data analysis. The first two themes identify diverse folk theories about either the text-to-image model's functioning, or its inner nature. The remaining two themes, instead, delve into the underlying processes and strategies that participants employed to build such theories. Table 2 summarizes the findings, providing a snapshot of the main takeaways of the study.

Table 2. Summary of the study findings.

	Themes	Sub-themes
Folk theories	**Mechanistic folk theories** Mechanistic theories see GenAI as operating as a mechanism, focusing on its parts and processes.	**GenAI as a pre-trained image-generator** - GenAI learns from a dataset of text-labeled images to recognize patterns. Then, it generates images based on user prompts using what it learned.
		GenAI as a search engine image-generator or as a search engine image-retriever - GenAI operates as a search engine either cross-referencing diverse retrieved data types and fusing them into a single image or retrieving pre-existing images in response to a textual prompt.
	Essentialist folk theories Essentialist theories see GenAI as an entity possessing specific qualities and abilities that are defined in relation to human characteristics.	**GenAI as human-like** - GenAI is an entity possessing human-like mental states, intentionally creating images.
		GenAI as human byproduct - GenAI is a product of human action, created, trained, and nurtured by humans, thus potentially inheriting human biases.
		GenAI as other-than-human - GenAI possesses alien-like characteristics, operating with its own unique mode of reasoning and interpretations of reality, consequently evoking unsettling emotions of mistrust.
Folk theorization	**Folk theorization process** The theorization unfolds through various paths, mostly progressively and through trial and error.	**Confirmatory additive process** - The theorization unfolds through a linear trajectory where theories are progressively consolidated either deductively, from broad hypothesis about the technology, or inductively, through local explanations prompted by specific images
		Circular iterative process - The theorization entails repeated formulations, verifications, and corrections of theories, whereby new data contradicting previous hypotheses may lead to theory evolution or crisis.
		Inhibited process - The theorization may be interrupted due to perceived difficulties in explaining particularly enigmatic cases.
	Folk theorization strategies The theorization process relies on a range of strategies aimed at supporting the interpretation of the model's behavior.	**Leveraging prior knowledge on AI** - Relying on pre-existing understanding of AI serves as a foundational explanation framework for participants with a technical background.
		Identifying with the GenAI - This strategy attempts to adopt the perspective of the model itself, to understand its "cognitive processes" and decision-making mechanisms.
		Drawing parallels with familiar technologies - This strategy entails comparisons between the GenAI and well-known technologies, like search engines and image manipulation software.
		Self-referring - This strategy implies referring to one's own experiences and practices and then applying them to the understanding of how the model might operate.
		Referring to humans - This strategy involves drawing parallelisms between human cognitive processes and abilities, like memory, and those attributed to the model.

By and large, individual differences played a major role in shaping participants' folk theories. The very same image or heatmap often led different participants to develop different theories, showing the relevance of their initial assumptions, previous knowledge, and way of reasoning in the theory generation process. However, essentialist theories, which mostly humanized the GenAI text-to-image model, were produced by the participants irrespective of their backgrounds and previous knowledge of the technology, signaling that the

technology itself, which create human-like outputs, may elicit humanness ascriptions. Although participants deviated more from the term used as a prompt when identifying the subject of representation for abstract images compared to concrete ones, their reactions did not significantly differ when the "correct" term was revealed. Rather, specific images containing certain "out of place" details (such as the mirror failing to accurately reflect its surroundings) elicited peculiar responses of unfamiliarity and bewilderment across participants, regardless of whether the images were concrete or abstract.

We have to notice that the same participants could develop different, even contrasting and contradictory, theories during the interviews, relying on different strategies and following different theorization processes. In fact, most of the participants engaged in building a variety of both "local explanations", to explain a particular output generated by the model, and "general theories", aimed at accounting the overall functioning or nature of the technology. In this context, heatmaps somehow helped several participants develop a general sense of the technology, but mostly generated doubts of its functioning, or were a stimulus to revise ongoing theories.

4.1 Mechanistic Folk Theories: GenAI as mechanism - How does the GenAI work?

In their endeavor to explain the behavior of the GenAI text-to-image model, several participants formulated a variety of mechanistic theories aimed at understanding the inner working of the model, namely, how it was able to create an image. This speculation involves considering its components, the criteria according to which it comes to a "decision", and/or the process it follows to operate. By and large, all these theories point to an idea of the model as a mechanism, which produces its outputs in response to user-provided inputs through a structured process resembling a pipeline. A prevailing notion among participants is that this pipeline implies the existence of a sort of database/dataset, seen as an extensive repository of data from which the model "learns" or "extracts" information.

The data analysis uncovered three primary mechanistic folk theories: the GenAI as a pre-trained image-generator; the GenAI as a search engine image-generator; and the GenAI as a search engine image-retriever.

4.1.1 GenAI as a pre-trained image-generator

The GenAI as pre-trained image-generator theory points to a mechanism that produces images through a two-phase process. The process first entails a training phase using a dataset of images annotated with specific labels: here, developers instruct the model to recognize patterns and specific features within the images,

thereby refining its capacity to associate visual content with textual information. Subsequently, the GenAI can generate new images based on the learned patterns and associations, triggered by a user prompt. This folk theory mainly originates from a subgroup of participants, specifically those with technical backgrounds, and those who had previous experience with GenAI. P1 well exemplifies this process by explaining:

> "a phase where numerous images are presented, and where the models are trained based on these images. Afterward, [a phase in which] it goes on to create something based on what it has learned from the provided images" (P1).

More precisely, a first element of speculation relating to this theory revolves around the creation of the dataset and how the GenAI "learns", which ranges from entirely human-driven to hybrid (humans-GenAI) and unsupervised techniques. For instance, participants like P17 and P19 conceived the dataset as a meticulously curated archive crafted by developers over the years, which manually associated labels and descriptions with the images: "*I imagine that there have been people who, for years, have communicated with this intelligence, cataloging numerous images. For an image with kittens, they said 'three kittens,'* [an image of] *a bed, 'this is a bed,' a landscape, 'this is a landscape,' with a whole series of descriptions to create an archive, let's say, of recognizable things"* (P17). Instead, others, like P6 and P5, conceptualized the dataset as created through a hybrid approach, which integrates supervised learning, where the GenAI is trained on pre-labeled images to grasp text-image associations, with unsupervised processes, in which the system autonomously acquires the ability to recognize the images, independently *"finding its own methods"* (P5) for labeling them.

A second element of theorization refers to how the model produces the images once the dataset is formed. Several participants highlighted that the GenAI may follow a statistical process, focusing on the frequency of label-image associations that might be present in the dataset: for example, if certain labels are often linked with specific colors, shapes, or objects in the dataset, the model becomes able to recognize and reproduce those visual elements once a term is given as an input. In this perspective, P1, while commenting on the "Wig", suggests that the dataset *"*[includes] *more images of 'Wig' where wigs are worn* [on humans]*, than wigs alone or on a mannequin* […] *probably by providing only photos of wigs on classic mannequins, those without faces, it would have probably just created an empty face".* According to some participants, another crucial factor influencing the GenAI's image generation is its ability to establish connections between words (and the corresponding images), forming a sort of *"thematic field"* connecting their semantic links. This thematic field

enables the model to contextualize the prompt. For example, when commenting on "Justice", P3 believes that the images is influenced by *"a set of words that can create a small context,* [...] *a context of justice, made up of a courtroom, an office, a scale, which together can perhaps create this idea of a thematic field.*"

A final point of speculation connects with the causes of perceived "errors" or "inaccuracies" in the model's productions. Participants found that textual-visual "associative learning" may lead the GenAI to errors due to casual associations, as the model may *"learn things incorrectly because they may happen to randomly occur many times*" (P1). An example is provided by P1, who reflects on the difficulty that GenAI faces in accurately representing hands:

> "Perhaps because hands move more than the face. I mean, the eyes, maybe the nose, and the shape of the head are read and are fixed features, while hands can assume many positions and are perhaps more diverse. So, it becomes more challenging to frame a standard idea of a hand" (P1).

Similarly, P12 explains the "*weird*" aesthetic of the "Mirror", where she notes the absence of a proper reflection and a focus on the mirror's "frame", in this way: "*if you've only been fed with images of framed mirrors, you might mistakenly think that the frame is the determining element* [of a mirror]."

4.1.2 GenAI as either a search engine image-generator or a search engine image-retriever

The theories of GenAI as a search engine conceptualizes the model as an engine that may either follow a multi-stage process directly operating on the images or simply retrieve pre-existing images from some sources.

The first conceptualization defines that the search process commences with the user's insertion of the prompt into the system, followed by the GenAI real-time processing of the user's request through a search in a repository, which can even be the entire Web. This process involves cross-referencing diverse types of data to identify and select the most relevant elements that may be associated with a certain subject given as prompt (e.g., "Anger"). This selection is then used as a basis to operate a synthesis, where those different elements are "fused" into a single image. P20 well summarizes how this theory works:

> "[GenAI] does a very quick search on everything he [the GenAI] finds on the web. It could be Google Images or even articles—basically anything where an association with anger is found and adjectives, definitions, nouns, everything associated with this word, [...] facial expressions or perhaps gestures, movements […] Then, he should condense these elements together, generating an image" (P20).

The participants primarily contributing to the development of this theory were those without a technical background or with a design background. Within this framework, they engaged in speculating about the nature of the repository, the process guiding the model in the search and synthesis, and their implications on the resulting images.

Firstly, participants formulated various hypotheses regarding the nature of the GenAI repository, which may be composed of a wide range of sources and data formats, like visual and textual materials coming from encyclopedias and news. Alternatively, the GenAI may conduct online searches exploiting well-known search engines, like Google or Google Images, which may explain why its productions align with the most prevalent views on the Internet, as highlighted by P20: "*Why this typology? Perhaps because it's the most developed type globally, so it found this configuration to be prevalent, or maybe it's not the most prevalent, but it's the most represented on the Internet.*"

Secondly, there was consensus that the GenAI possesses the capability to *"cross-reference all existing documents, photos, videos, and texts"* (P14), analyzing heterogeneous and extensive volumes of data during the search. The synthesis, instead, was seen mainly as a gradual phenomenon, where the construction of the image was explained in terms of a progressive incorporation of elements from a starting point. P18, for example:

> "It clearly starts from where it has more material, so the redder it is, the more likely it found material on that particular thing. […] I think it started more or less from this area and then gradually moved further outward" (P18).

The theory of GenAI as a search engine image-retriever, instead, portrays the model as a mechanism for retrieving pre-existing images. This kind of theory was formulated only by individuals with neither a technical nor a design background, and no previous experience of GenAI. In this perspective, when a term is given as prompt, it is straightforwardly connected with an existing visual representation in the repository, which is provided as a result. P8, for instance, says that:

> "In my opinion, there's an archive of billions of images, you say 'steering wheel,' and it fetches it. [...] When you refer to something where the semiological relationship is direct, like a chair, a table, a steering wheel… this is undoubtedly one of the images in the database" (P8).

The three mechanistic theories that we identified showcase both similarities and differences. A primary point of difference relates to where the participants position most of the model's work in the generation process. The first theory suggests that the bulk of GenAI's workload lies in training and learning from a dataset, whereby the work performed between the prompt and the image generation is simply a recall of what it has previously learned. By contrast, the second and third theories conceive the GenAI's work primarily located after the user's input, involving a search activity that may require strong computational efforts. Another crucial aspect is the nature of the returned image: while the first two theories posit that the image is newly generated, the third one states that it is a pre-existing image simply retrieved.

4.2 Essentialist Folk Theories: GenAI as intentional entity - What is the nature of the GenAI?

On many occasions, participants, irrespective of their backgrounds and previous knowledge of GenAI, adopted a more "essentialist" stance, developing theories that point to the "nature" or "essence" of the model. Here, the text-to-image model is meant as an entity with particular qualities and abilities that may cause its behavioral manifestation (i.e., the production of images). In fact, participants saw the source of this causality in a supposedly "intentionality" owned by the GenAI, thus attributing a variety of "mental states" to it. This kind of theorization ultimately takes as a point of reference the human being.

We identified three distinct essentialist theories that depict the model as an intentional entity. The first theorization regards the GenAI as a proper human-like agent; the second one sees the GenAI as a human byproduct; and the last one conceptualizes the GenAI as an other-than-human entity.

4.2.1 GenAI as human-like

The folk theory of GenAI as human-like conceptualizes it as an entity endowed with anthropomorphic characteristics, moved by an intentional purpose. This is visible in the fact that participants, including those with a technical background, recurrently addressed the GenAI with personal pronouns, namely "lui" (he) and "lei" (she), which in Italian are only used to refer to humans.

In the theory of GenAI as human-like, a primary attribute ascribed to the model is the ability to create something in a "thoughtful" manner, being able to properly understand the concepts it represents. Several participants described the generation process as an activity guided by conscious thoughts. P16, for instance, explained the heatmap of "Love" as a trace of the GenAI's retrospective thought: *"Love is something you*

share, I mean, you need to be two […] [P16 points at the heat map] *she is aware that typically it involves two people.* […] *But I can't understand why she removed the person, I mean, based on what criteria she removed it.* […] *So maybe she imagined it and didn't... I mean, did she hesitate? Did she remove it? Did she remain unsure until the end whether to include it or not? Did it become a heart?"* (P16); while P1, about the same image, says *"It probably imagines that a human being is enough to convey this idea, perhaps it doesn't grasp the concept that love involves two individuals, that is, the fact that it's a feeling from one person to another*."

Another key characteristic that certain participants ascribed to the model to explain its behavior relates to its purposefulness in the process of image generation, seeing it engaged in decision-making processes and directed toward specific human-like objectives. An illustrative example is shown by P14, who attributes to the GenAI the intention to satisfy humans:

> "If the goal of artificial intelligence was to create an image that could unite, that could bring together more people, then it certainly could not represent cases of private or public justice because there are so many cases. This is an image that, on the other hand, everyone can agree upon. I think the software tries to provide an answer that can satisfy as many people as possible" (P14).

A last characteristic that participants attributed to the GenAI is the possession of "personal" opinions and moral judgments, then expressed through the generation process: *"I see that artificial intelligence thinks that a person who is alone is a person in love"*, P19 says. Furthermore, participants ascribed to the GenAI's behavior proper political and social attitudes to explain its products, like GenAI as "fascist" in relation to "Anarchy" (P10, P12) and "body-shaming" regarding "Laziness" (P4, P20). A direct consequence of these attributions is to see the images as "deliberate acts", provoking reactions of agreement with or opposition to the GenAI, for acts that were perceived as morally "right" or "wrong" compared to the participants' values: for example, *"Well done AI, this time you're right! For once*" P6 says, and "*A woman to represent laziness... Why a woman in the first place? Asshole*!", P12 remarks.

By sharing fundamental characteristics and abilities of human beings, the theory of GenAI as human-like is quite reassuring for the participants. It states that the GenAI has beliefs, goals, and values similar to those of human beings, and thus can be criticized or praised, but ultimately represents a sort of familiar entity.

4.2.2 GenAI as human byproduct

A second essentialist folk theory that we identified refers to GenAI as a human byproduct, seeing it as a sort of sentient (or semi-sentient) entity created, trained, and nurtured by human beings, a *"product of human action"* (P9). Being trained according to human instructions and processing data generated by humans, the model appears to be endowed with the capacity to emulate, mirror, and faithfully reproduce human's collective imagination and reasoning, which may explain why certain images resonate with the ways the participants imagine things and events: "*it relies on a mode of observation that is inherently human. In other words, it studies the human way of observing, and consequently constructs images in that same manner*" (P7).

The theory has three fundamental features. The first one points to the "subordination" of the GenAI, its dependence on humans who "feed" it. Several participants used the metaphor of feeding, emphasizing how the GenAI is an entity to whom humans provide data-as-food, which is essential for its functioning. For instance, P12 invokes the classic behaviorist theory to explain how humans conditioned the model, making it learn through the repeated association of stimuli and responses:

> "I think artificial intelligence consumes and develops based on what we feed it in an almost Pavlovian manner. If we repeatedly associate a stimulus with an outcome, even if that stimulus has nothing to do with the outcome, like the bell and the meat, the intelligence will react to that stimulus with that particular outcome through association, through associative learning. […] Because we do the same. Why are we afraid of lightning? Why do we associate lightning with fire? We associate lightning with cauterization. We have learned through association that lightning is connected to fire, so we instinctively avoid lightning" (P12).

Not only the GenAI may thus be similar to us, because it undergoes the same learning procedures, but it is also subordinated to us as it relies on human guidance and control. In this sense, P19 connects the idea of subordination to the impossibility of blaming the GenAI for its products, because it simply follows human instructions, being the model a sort of "diminished" human:

> "Despite being a learning machine, the model is constructed by us, so the machine does what we instruct it to do and trains with images and data that we have generated. […] And, of course, this work

is done through rules that are provided, so maybe, poor artificial intelligence, it's not to blame; we told it to do the wrong thing" (P19).

A second feature of the theory is "reproduction", which attempts to explain why certain images seem to convey the same imaginary that we also have. Here, the model may replicate the elements that it finds in our imagination, whereby humans shape a vision of reality through their iconographic representations, as in the movie imaginary or in the iconography of advertising. However, as noted by P19, the reproduction concerns humans' constructions, not the reality itself: "*Artificial intelligence reproduces an imaginary world and values according to which we have constructed this society,* […] *but, honestly, they do not correspond to reality*" (P19).

The third characteristic also points at explaining the biases that participants observe in the generated images, and it connects with the idea of "inheritance". The GenAI inherits the biases of those who nurtured it, reflecting their common conceptions and perceptions, as well as their cultural perspective, which is particularly embedded in the data that are fed to it. For example, P16, says that *"in the images I see this American imagery. Since these artificial intelligences are often produced in the United States, in Silicon Valley, etc., rightly, the data is taken from there"*. Within this framework, this replicative entity is seen to amplify hegemonic thinking, reflecting mainstream beliefs and weltanschauungs, as P18 articulates:

> "Because historically, in recent times, it's possible that the conception of anarchism, which is generally portrayed by the mainstream media, is seen as rebellious, truly rebellious, and violent […] since we live in a period where the far-right or the right, in general, has a more significant influence on information than the left, culturally, it shapes the image of an attitude considered by most as wrong or not in line with the prevailing ideology. […] Because artificial intelligence is fed by human beings, and therefore, unfortunately, it is more prevalent to find a right-leaning thought than a left-leaning one in the last ten years" (P18).

The theory of GenAI as human byproduct is certainly reassuring in these participants' eyes, as it gives the control of the technology back to its creators. While it grants the GenAI certain abilities that are typical to humans, it substantially denies autonomy, somehow taming an underlying fear of a possible "rebellion" of an entity that may overcome us in terms of efficiency or may not align with our goals and values.

4.2.3 GenAI as other-than-human

The folk theory of GenAI as other-than-human conceptualizes it as akin to an organism with alien-like characteristics, an entity possessing its own unique mode of reasoning and personal codes for interpreting and analyzing reality, fundamentally distinct from humans. Participants often describe this entity in terms of what it lacks (i.e., characteristics it does not possess), presenting it as obscure and elusive, like a being whose "mental" and decision-making processes are inaccessible to human explanations and understanding. Such a viewpoint becomes particularly salient when participants struggle to explain the model's behavior that produces results that completely contradict their expectations and previous theorizations, as highlighted by P17, "[The GenAI considers] *exactly the opposite of what I was saying. That's because he is an artificial intelligence, and I am a human being. Which means we think in two different ways"* (P17). In these situations, conceptualizing the GenAI as an alien entity may be a sort of last resort before giving up on trying to explain the unexplainable: here, the GenAI is perceived as an intentional, intelligent, but alien entity operating with its own kind of reasoning and goals.

Despite its alien nature, several participants described its characteristics, in the attempt to make sense of the unknown. For example, they focused on its unique sensory apparatus, especially the visual sense, used for observing and interpreting the world. The GenAI has a way of "*segmenting reality*" different from that of the humans, as highlighted by P2:

> "In my opinion, its rule is that everything is much more divided into elements than how a human does it. […] and especially the elements in which an AI divides are elements that don't make sense for a human being. I mean, for me, the steering wheel is an element, the symbol is an element, the joints not so much, but evidently for an AI they are not" (P2).

In fact, despite the participants' attempts to explain its nature, the GenAI remained fundamentally obscure for them, provoking concerns, unsettling feelings, and mistrust towards the technology: conceiving the model as completely alien and unexplainable, yet intentional and intelligent as humans are, entails the possibility of its independence and misalignment from human values and goals, leaving participants with a sense of "unfamiliarity" despite their attempts of taming the unknown through theorization:

> "I find this whole AI thing a bit scary because I see it as a means that, okay, it's useful, but potentially it could also gain a certain type of independence. And so, it's not subject to control […]. Independence in the sense that it could also enact choices that have not been controlled by humans" (P7).

4.3 Folk theorization processes

Participants engaged in theorization activities following different trajectories. Several theories were formulated through a gradual linear process, while others followed a circular path, made up of iterative theorization, assessment, and reformulation. Often participants alternated between "local" explanations, namely, attempts to figure out why a specific prompt originated a particular image or why a certain heatmap had the shape it had, and more general theories addressing the overall functioning or nature of the GenAI.

4.3.1 Confirmatory additive process

The confirmatory additive process entails an approach in which the formulation and consolidation of a theory are carried out progressively, along a linear trajectory, where each "step" of the theory is built on the previous one. Within this process, local and global explanations complement each other, whereby the participants follow two primary paths for theory building, a deductive approach and an inductive approach.

In the deductive approach, broad pre-existing theories or initial general hypotheses, often constructed on their preliminary or previous understanding of the GenAI, are progressively detailed and confirmed through their application to singular cases. For instance, P1 initially explained the text-to-image model as a system that learns to recognize certain patterns in the dataset and then generates novel products grounded on what it learned. As P1 delved into the exploration of the images and the heatmaps, her initial hypothesis turned into a more detailed theory. For example, she identified the presumed criteria that the model may use to detect patterns across the dataset and explained the heatmaps' appearance: the redder an element is in the heatmap, the more frequently it appears in the dataset associated with the term given as a prompt. In other words, after having established a general causal relationship between the content of the dataset and the generated outputs, she started theorizing about the composition of the dataset and how it was used to generate the images, gradually enriching and consolidating the theory.

Conversely, other participants adopted an inductive approach. In these cases, initial theoretical formulations emerged locally, in response to the provided stimuli, and only after a general theory was formulated. For

example, P20 initially expressed only a limited understanding of the model's ability to transform input into output. As more images were presented to her, she formulated a more general theory about the GenAI's behavior, stating that the GenAI gradually creates the images, like an "*expansion of an oil stain, progressing incrementally*" (P20). She then hypothesized that this process originated from the data that the GenAI could find on the Internet, eventually detailing the different stages that the model follows in the generation process, from the initial Internet search to the gradual condensation of the different retrieved images into one. To validate the theory, she assessed its applicability to individual cases, also going back to the images that were presented in a previous moment. In this process, anomalies or deviations (e.g., images that cannot be logically connected with their prompt, or containing unfamiliar details) are mostly assimilated and explained, even forcibly, in terms of the theory being developed: here, "novel data" are more meant to offer confirmation to the forming beliefs than to provide occasions for revising the hypotheses.

4.3.2 Circular iterative process

The circular iterative process unfolds through repeated formulations, verifications, and corrections. This cycle becomes apparent when participants encounter cases that do not fit in their theory. This may lead to the evolution of the theory, entailing the integration of new explanations in the theoretical "corpus" or the complete revision of its statements. Alternatively, the current theory may be discharged, and a new theory may be formulated. Within this process, disconfirming cases are perceived as stimuli for further sense-making. Again, we can see here a deductive and an inductive approach.

In the circular iterative deductive approach, an existing global explanation undergoes iterative testing to validate its applicability to local cases. Differently from the confirmatory additive process, here participants are engaged in frequently doubting and questioning their initial hypotheses, especially when a case appears to contradict them. When this happens, they carefully assess what aspects of the theory need to be discharged or adjusted and what aspects can be confirmed.

For example, P7 developed a general explanation of GenAI as a human byproduct: she explained that she and the GenAI deemed the same image aspects significant because, *"simply put, it relies on a mode of observation that is inherently human* […] *and consequently constructs the images in that same manner".* However, this initial theory was disrupted when P7 encountered a series of cases that contrasted it. She then recognized that she was mistakenly giving for granted that both the entities (she and the GenAI) were

characterized by the same kind of reasoning. This led her to gradually reconceptualize the GenAI as an entity substantially different from humans. However, later on, she went back to the initial general theory, swinging between the two main conceptualizations: on the one side, she acknowledged "*that the AI draws from an imaginary that is human, so it will reflect the majority* […] *of people*". On the other hand, she considered the GenAI *"as a means that, okay, it's useful, but potentially it could also gain a certain type of independence* […] *in the sense that it could also enact choices that have not been controlled by humans"*. This continuous oscillation between different theories signals the participant's constant attempts to make sense of a system that is obscure, whereby to account for all the "anomalies" it is sometimes necessary to change the theoretical framework itself.

Instead, in the circular iterative inductive approach participants mainly propose local explanations, generalizing them to a global theory only when several recurrences are identified. However, if such theory is not applicable to subsequent cases, they temporarily set it aside in favor of more suitable, on-the-spot circumscribed conceptualizations. In these cases, the theorization process assumes a flexible trajectory, replacing general and local theories as new cues emerge.

P13, for instance, did not have initial insights about the functioning of the model: she formulated only local explanations about why a particular image had the appearance it had. However, the exposure to a series of photo-like images led her to speculate that GenAI might retrieve pre-existing pictures from some kind of repository. When she noticed the absence of reflection in the "Mirror", she started thinking that the GenAI could actually generate novel creations by collecting and blending diverse information. Nonetheless, other subsequent photorealistic images made her reconsider even this hypothesis, leading her back to the original idea of GenAI as a search engine image-retriever. Then, she developed the theory further, by explaining that the GenAI detects "*in percentage everything that is within the average*" and retrieves the most representative image according to the prompt. She then went back again to the idea of GenAI as an image-generator search engine, moving back and forth as long as other photos or heatmaps were presented. Towards the end of the interview, she ultimately stated: *"In most cases, they seem like images created ad hoc, and at this point, perhaps I can recognize this modus operandi to some extent. In a few cases, they seem like images taken from a photo album"*, thus reconcealing both the theories and acknowledging that the GenAI could activate either process as needed.

4.3.3 Inhibited process

The inhibited process encompasses a temporary interruption of theorization due to perceived difficulties in explaining particularly enigmatic cases, which are ultimately considered inexplicable. For example, P13 states that *"I don't know. I just can't find a meaning or try to reconstruct a path that could have been taken, even if it was just logically and not necessarily rationally, let's say. No, I really can't understand".* In certain cases, the difficulty in articulating a reasonable explanation led participants to attribute the GenAI's output to a glitch or system error, underscoring its technical limitations. For instance, P6: *"No! I dissociate myself, let's move on. I'm not going to articulate a thought on these things. I can't explain the process. I can't understand it. [...] In my opinion, it's a mistake*". In other cases, participants attributed their inability to explain the GenAI's behavior to their own human nature: *"I try to give a human explanation to what the AI does, but there isn't one, you understand? So, it's pointless to come up with explanations."* (P2). In all these cases, the theory building process was interrupted and participants often experienced frustration and unsettling feelings for the inexplicability and unfamiliarity of the GenAI and of its outputs. However, even though the folk theory process was inhibited, the participants could not help but make some sense of these cases, categorizing them either as machine errors or as phenomena that went beyond their limited understanding.

4.4 Folk theorization strategies

We observed that participants relied on various strategies to theorize about the GenAI. These strategies were mainly aimed to trace certain unexplainable GenAI's behaviors back to more familiar reference points.

4.4.1 Leveraging prior knowledge of AI

To formulate theories about the functioning and nature of the text-to-image model, certain participants exploited their pre-existing knowledge of AI. On the one hand, this strategy served as a basis to formulate initial hypotheses on the GenAI's inner working principles for participants with a limited knowledge of AI. On the other hand, it supported participants with a technical background to explore not-yet-known features of the text-to-image model.

In the first case, participants with a basic understanding of AI referred to news sources and past informal conversations to formulate general descriptions about the GenAI's behavior, like *"intelligences that retrieve*

information from databases" (P16), thus providing a starting point for theory building. In the second case, participants with more detailed knowledge of AI, used their understanding as a means to establish analogies or ground their subsequent explanation attempts. For example, P15 explains that "*what I know is that these systems have an input part, a series of data that are entered to train this artificial intelligence* […] *like classified datasets that allow it to imagine. This process of imagination, however, happens through a sort of black box*". Similarly, P2 and P3 referred to the features of convolutional neural networks, a specific type of neural networks mainly used for image recognition and classification, to suggest that the GenAI might follow analogous processes, especially in analyzing and segmenting elements within the images.

Nevertheless, somewhat paradoxically, participants with a more in-depth previous knowledge of AI faced more challenges in confidently formulating detailed mechanistic theories about the inner working principles of the model. This arose from the acknowledgment of both the inherent opacity of the GenAI and the perceived lack of expertise in that specific GenAI model, which led them to be more cautious and somehow doubtful. For instance, P2, says "*surely there are many photos of mirrors taken in this way, to represent mirrors* […] *probably it was trained on them. However, I don't know which AI we're talking about, which generative model*." In these cases, it appeared that the awareness of the inner complexity of the GenAI hindered the development of detailed mechanistic theories.

4.4.2 Identifying with the GenAI

A second strategy adopted by the participants to formulate folk theories was to try to *"put themselves in the AI's shoes"* (P6). Overall, participants' endeavors leaned towards the "dehumanization" of their own human reasoning process to embrace the supposed GenAI's viewpoint through a practice of "imagination". For instance, P17 regularly considered *"how an artificial intelligence would think*":

> "If I were an artificial intelligence, I would tally how many times the word 'memory' has reappeared, perhaps taking the most frequently repeated case, so [I would recognize that] people identify that as memory, and I would create an image that could be a combination of those" (P17).

This strategy was often used in cases of violated expectations, whereby the participants attempted to identify the differences between their interpretative codes and those of the GenAI, as P2 explains:

> "Now I'm starting to think a bit more like the AI. [...] because after I realized my mistake, I started thinking a bit differently, more in terms of how the training set could be done, right? […] to go halfway between how a human might think and what the AI might have considered important in the images of the training set."

In general, even though the participants were well aware of the impossibility of genuinely adopting the GenAI's perspective, this strategy played a dual role. On the one hand, it facilitated the articulation of certain conceptualizations about the GenAI, serving as a sort of metaphor for explaining concepts that appeared extremely technical to certain participants. On the other hand, it allowed participants to temporarily distance themselves from their "personal perspective", enabling them to consider alternative explanations.

4.4.3 Drawing parallels with familiar technologies

Another strategy that participants used concerned referring to well-known technologies, such as search engines and image creation software. This strategy enabled participants to elucidate the GenAI's behavior within a more familiar frame.

In most cases, participants made comparisons between the GenAI and search engines, since both technologies allow users to input a text and receive a result, on the basis of an analytical process conducted on certain data. Other participants, especially those with a design background, found analogies between the GenAI and image manipulation software. For instance, P18 often drew comparisons between the model and a rendering engine, a software for generating photorealistic images of three-dimensional virtual models: *"it seems from this perspective as if it were a rendering engine you're talking with because, in fact, the more specific you become not only in describing the content but also in describing how this content should be generated, the closer you get to that solution"*. Moreover, the comparison with image editing software allowed P17 to conceptualize the image generation process as a photomontage: "*Once it has created this archive, when you make a request with a phrase, I imagine that it gathers all the information it can link through the words to its image archive and tries to Photoshop them together*".

In sum, making analogies with familiar technologies enabled participants to increase their perceived comprehension of the model, by offering a familiar conceptual framework that facilitated their understanding of unfamiliar processes.

4.4.4 Self-referring

On many occasions, participants also employed a strategy of self-referring, drawing upon their personal practices and experiences to explain the GenAI. For instance, several participants with a design background used their design practice as a point of reference to elucidate how the model translates text into images. P7, for example:

> "I imagine it in a similar way, like when they tell me, 'design me this object, design a new steering wheel.' Okay, the first thing I do is to make a survey... […] I open Google Images, for instance […] and I look for what, for others, is the definition of a steering wheel. Then maybe, you know, there are things that interest me because I add my touch, so I make choices, set criteria, and choose." (P7)

On other occasions, participants connected the GenAI's behavior and outcomes with their personal experiences. For instance, P17 equated her own peculiar ways of thinking in that particular moment to that of the GenAI: "*In this moment, I am reasoning about the elements that define something, so if I reason in this way, I imagine that it would reason similarly. So, there's also the mountain because a mountain can make you understand that you are, for instance, in the mountains*."

Essentially, the self-referring strategy employed by these participants first provides insights into the individual's "self", through a recounting of their experiences and practices, and then uses them to interpret the underlying mechanisms of the GenAI.

4.4.5 Referring to humans

Participants, on several occasions, also linked the GenAI to generic "human beings", especially to develop essentialist theories about its nature. In so doing, they described the GenAI on the basis of what they thought characterizes a "human being", or, more simply "us". For example, P18 drew a parallel between human cognitive processes, which he considers often operating below the threshold of conscious awareness, and those of GenAI, characterized by a sort of "black boxness":

> "There are certain things that are so inherent to us, so ingrained, that we don't even know they are coming to mind. So, I don't think our process of constructing images is easily representable or schematizable" (P18).

Alternatively, participants started from supposedly processes shared by humans and the GenAI, like remembering, to identify differences between them, as P16 remarked when talking about human and GenAI reasoning: "it's like... The difference between a city you live in and a city on the map. The city you live in is the city of your experiences. The city on the map is made of lines, outlines, fillings".

Finally, a minority of participants referred to humans to offer a "negative" example of what the model is not able to do, stating, for example, that it does not have feelings, memories, or sensations, which, instead, are inherently possessed by human beings. This helped them delineate the boundaries between what is peculiar to humans and what pertains to GenAI, while also allowing them to devalue the technology by highlighting what it is not capable of doing: in this way, they mitigated the unsettling feelings that may come from an "entity" that potentially has more-than-human abilities (e.g., creating images faster than humans).

Overall, this strategy facilitated the development of theories about the nature of the GenAI, allowing participants to identify similarities and differences between it and human beings and to articulate their conceptualization more easily, through a conceptual frame that was more familiar to them.

5 DISCUSSION

The study findings give an answer to the research questions that drove our research.

In response to the first research question (*What kind of folk theories do people develop about a GenAI text-to-image model starting from its outputs?*) we identified the two main types of folk theories: those emphasizing the mechanistic aspects of the model, focusing on its functioning and operations, and those addressing its inner nature, thus adopting an essentialist stance. As for the mechanistic theories, participants conceptualized the GenAI text-to-image model as a mechanism, the behavior thereof is structured around a sort of pipeline: the model follows a series of precise steps either to generate an image in response to a textual prompt, or to retrieve an existing image from an extensive repository of data. Conversely, essentialist theories conceptualize the text-to-image model as an entity endowed with intentionality and agency, connecting it to the realm of humans: in this perspective, the text-to-image model may be something that is similar or opposite to human beings, or a mere human creation, thus relegated to a subordinate position.

As for the second research question (*How are these theories formed, and how does the folk theory development process unfold over time?*), our findings show that theorization predominantly unfolds through trial and error. Participants followed different paths to build their own folk theories, pursuing either a linear

additive process, where each step of theory development is grounded on the previous one, or a circular iterative process, in which phases of theory building are alternated with phases of assessment and theory refinement or substitution. In so doing, participants relied on a variety of strategies to explain the technology's behavior, ranging from referring to previous AI systems and more familiar technologies, to connecting with their own practices and experiences and their knowledge of human beings. Nevertheless, there were cases when these trajectories of theorization were interrupted due to the inexplicability of the text-to-image model. Despite these interruptions, participants strenuously pursued continuous sense-making to face the opaqueness of the technology.

In this sense, we made three substantial contributions providing three critical insights: i) we identified the essentialist theories, a peculiar kind of folk theories that emphasize the human-like qualities of the technology, which stem from the fundamental black box nature of the text-to-image models and partially provide reassurance from their unfamiliar nature; ii) we offered a nuanced account of how folk theories are developed, demonstrating the remarkable malleability of their construction: this flexibility may likely stem from the peculiar nature of the technology, which seems to encourage people to produce a variety of theories and continuously refine them in an attempt to explain its outputs - initially appearing straightforward, but ultimately alien upon closer examination; iii) we highlighted the role of individual differences in the theory-building process, emphasizing that the person's background may influence the generation of folk theories: this underscores that one-size-fits all approaches to modifying such theories may not be effective, as individuals with different backgrounds and experiences may interpret the technology's behavior in diverse ways. Finally, as a supplementary fourth insight, we point out the people's fundamental need to make sense of a technology that performs tasks typically associated with humans, thereby naturally evoking attributions of human-like abilities, which, nonetheless, cannot be explained in human terms.

5.1 Essentialist theories: Anthropomorphizing the text-to-image model to find reassurance against its unfamiliar nature

As a first key contribution, we identify the essentialist theories, which focus on the "essence" of the text-to-image model, rather than on its mechanisms and ways of functioning. This kind of theories quite differ from those pointed out in research about folk theories of social media algorithms. In such research, the theories outlined range from those attempting to explain specific operations of the algorithm (DeVito et al., 2017;

DeVito, 2021; Grill & Andalibi, 2022; Karizat et al., 2021), which correspond to the mechanistic theories we identified, to abstract theories that point to more general explanations about the technology, meant as "something" that may cause something else (DeVito et al., 2017; DeVito, 2021).

By contrast, the essentialist theories attribute intentionality and agency to the machine and take the human being as a point of reference to understand its "essence" and abilities. In this sense, they also differ from the artists' folk theories of text-to-image models described by Shelby et al. (2024). Such theories concern *the use* of the GenAI models in creative practices, seeing them as tools or media, rather than as "entities" possessing human-like features. The essentialist theories we outlined, therefore, may be seen as a peculiar kind of folk theory, one that is specific to how text-to-image models may be perceived as something more than "mere tools".

It is true that attributing human-like features to technology is a well-known phenomenon in HCI research: not only users may humanize technological artifacts (e.g., Candello et al., 2017; Rapp et al., 2021), as also emphasized by Computers Are Social Actors (CASA) theory, which suggests that people may apply human social rules and expectations when interacting with computers (Lee & Nass, 2010); but also, designers may intentionally reproduce fundamental aspects of being human in their designs (e.g., Lee et al., 2019; Følstad et al., 2023). Siles et al. (2020) noticed that people may conceptualize Spotify as a person-like being that engages in surveillance to provide music recommendations, also using pronouns like "he" to refer to it, which are typically employed with human actors. This tendency can be also seen in our study, where participants sometimes implicitly conceptualized the text-to-image GenAI as a human-like entity, by referring to it with "human" personal pronouns.

However, humanization seems more pronounced in the context of text-to-image GenAI, where theories about the nature of the technology, meant as an intentional agent, represent a large portion of the theories developed by the participants. More importantly, these conceptualizations arise even in the absence of a direct interaction with the GenAI text-to-image model, as the participants were only shown the images it created. Moreover, not only participants anthropomorphized the model, ascribing human-like "mental states" to it. They also saw it as a human byproduct, which, on the one side, shares the same abilities and objectives of its creators, and, on the other side, it is in an inferior being relegated to a lower hierarchical position. This practice of infrahumanization, where people may treat infrahumanized targets as entities lacking certain human

characteristics (like autonomy) but not others (like competence) (Leyens et al., 2003), may also be encountered in user interactions with chatbots, where, nonetheless, infrahumanization is mostly implicit, traceable, for example, in how people address the chatbot when it is not able to satisfy their needs (e.g., by using an imperative tone or by devaluating its abilities) (Rapp et al., 2023a; Rapp et al., 2023b).

In this study, instead, we have seen that participants explicitly theorize the text-to-image GenAI as a less-than-human entity, a sort of "Frankenstein", endowed with human abilities that may explain why it is able to create human-like products, yet not autonomous and meant to serve the human goals. This kind of explanation seems to be peculiar to technologies that effectively produce outputs that are extremely similar to those produced by humans, and thus may need to be explained by referring to human beings, but in which their extreme efficiency and "superhuman" skills need to be balanced by denying other human qualities (like autonomy) that may put people at risk, taming in this way an underlying fear of "rebellion" of the technology.

On the same line, the theory of GenAI as other-than-human also appears peculiar to text-to-image models. Here, the technology is conceptualized as an intentional, intelligent being, which, nonetheless, remains completely obscure. Here, despite a human or superhuman level of intelligence, the text-to-image GenAI is thought so different that its abilities can only be explained through an act of imagination, by identifying with the model, or cannot be explained at all. If we compare this conceptualization with those identified in previous research on folk theories, it appears rather unique, likely stemming from the fundamental black box nature of the technology, which, nevertheless, is able to create products similar to those produced by humans. Participants appear to rely on this theory when all the other explanations seem to fail, and when even the theory of GenAI as human byproduct is not able to account for the "anomalies" of its outputs: how is it possible to believe that the technology has no autonomy and precisely follows what we tell it to do, if we are not able to understand at all some of its outputs?

In this context, the theory of GenAI as other-than-human is the last resort before giving up on trying to explain the unexplainable. However, this theory leaves participants with unsettling feelings and mistrust towards the technology. The theory somehow entails that the text-to-image GenAI could pursue its own goals and values, which, nevertheless, remain inaccessible to us, creating space for acts of independence that may go against our own interests as human beings. Despite evoking eerie feelings, theorizing the technology as an alien entity can still be more reassuring than facing the unknown: no matter how alien it can be, but a

technology that is seen as an intelligent and intentional agent remains something one can potentially "relate" to rationally, which is much more reassuring than thinking of GenAI as something that cannot be compared to human beings at all.

5.2 The malleability of theory construction: proceeding through trial and error to explain the obscure nature of the model

As a second key contribution, we offer a nuanced perspective on how people form folk theories about text-to-image technology, describing the theory formation process as highly flexible, characterized by continuous trial and error, as well as by theory abandonment and refinement, where individuals rely on different strategies and follow diverse paths to shape their conceptualizations.

Differently from previous research (DeVito et al., 2017; Siles et al., 2020; DeVito, 2021; Grill & Andalibi, 2022; Karizat et al., 2021), we did not ask participants to recount the theories that they already had about the technology: rather we observed "in person" the theory formation process, even from its initial stages and as "new information" (i.e., new GenAI outputs) was introduced. Actually, previous research only seldom engaged in the exploration of folk theory formation process: when this process came as a matter of interest, research highlighted that people may draw from multiple and heterogeneous sources of information for developing a theory; moreover, it emphasized that theory formation may be purely additive or may require confirmation, also noticing that folk theories may be malleable, thus changing as new information is provided (DeVito et al., 2018; DeVito, 2021).

The study findings confirm the preliminary insights provided by such research, adding further details about the theory building process. We pointed out that people may follow either a confirmatory additive path, where anomalies and violated expectations are often reinterpreted to fit into the theory at hand, or a circular iterative path, where they may swing between different theories, and disconfirmatory evidence may be used to develop or change the theory in progress. In so doing, we highlighted that individuals may rely on inductive or deductive approaches and also build a variety of local explanations or general theories, which may even be contrasting or contradictory. Not only were these trajectories not identified in existing research, but these findings also demonstrate the remarkable malleability of folk theory construction, far surpassing what has been observed in previous work (DeVito et al., 2018; DeVito, 2021). The extreme malleability of both the theory formation process and the theories produced might be due to the specific nature of the technology, being

opaque and obscure, yet performing tasks that are similar to those performed by humans: the clash between the familiarity of a human-like activity, like creating images, and the unfamiliarity of its inner functioning, may encourage people to produce a variety of theories and continuously refine them, in the attempt to account for what seems to be explainable at first sight, but at a closer look may reveal its alien nature. This finding may have important implications for XAI research: as folk theories are extremely malleable, they could be potentially steered towards certain direction by using XAI methods.

Furthermore, we identified a variety of strategies that people employ to develop their folk theories. These strategies extend what has been found by previous research, which noticed that people may rely on their knowledge and assumptions about how the technology work (DeVito et al., 2018). These study findings, instead, point out that individuals not only refer to their previous understanding of AI, but also refer to technologies that they believe are similar to the text-to-image GenAI. Furthermore, people may ground their conceptualizations on the knowledge they have of human beings and even to their own experiences and practices.

5.3 The role of individual differences in folk theory generation: people with diverse backgrounds and levels of previous experience with GenAI produce different theories

As a third contribution, we point out the role of individual differences in the theory construction process, highlighting that people with different backgrounds and levels of previous experience with GenAI/AI conceptualize text-to-image models differently. In fact, we recruited individuals with diverse backgrounds, even involving people who did not know the technology and never interacted with a text-to-image model. Previous research, instead, focused on specific users of the technology under investigation, thus recruiting more homogeneous samples (e.g., DeVito et al., 2017; Siles et al., 2020; DeVito et al., 2018; DeVito, 2021; Grill & Andalibi, 2022; Karizat et al., 2021; Shelby, 2024): this homogeneity hindered the ability to identify the personal factors that may influence the theory formation process, as well as the variability of the theories formed across different individuals.

Instead, our approach allowed us to outline differences in the theories formed and the strategies used for their formation across participants with different backgrounds. Participants with a technical background and/or prior experience of (Gen)AI exhibited greater uncertainty and caution in formulating hypotheses on the inner functioning of the text-to-image model, being aware of the intrinsic complexity of the technology. Moreover,

they tended to align their explanations with the functioning of other, even pre-generative, AI systems. By contrast, participants with no technical background progressively demonstrated more confidence in formulating mechanistic theories. They referred to familiar non-AI-based technologies, such as search engines and image manipulation software, ultimately developing somewhat more naïve theories that perceive GenAI as a kind of search engine. However, irrespective of their backgrounds, all participants produced, both explicitly or implicitly, theories that see the model as an intentional entity, showing a tendency that may be independent of one's previous background and knowledge about the technology, and may thus be retraced to the qualities of the technology itself.

The importance of the person's background in generating the theories, however, is also testified by the fact that participants often relied on their personal experience and practices to explain the text-to-image GenAI. This, again, might be favored by the fact that the model performs tasks that appear extremely similar to those carried out by human actors, so that they are encouraged to make comparisons with themselves. It comes as no surprise, therefore, that people with a design background connected the text-to-image GenAI with their own design practice, where they have to create images by their own, following a certain creative process. These findings may be important if we aim to modify people's folk theories about the text-to-image GenAI: they highlight that a one-size-fits all approach may not be completely effective, as individuals with different backgrounds and experiences may tend to interpret the technology's behavior differently.

This said, a further way to tie the finding together is to connect them with a key theme that emerged across the participants' accounts and revolves around the search for meaning when faced with a potentially nonsensical phenomenon.

5.4 Finding meaning in the unexplainable

The number and variety of local explanations and general theories developed during the study by the participants express the people's fundamental need to make sense of the unexplainable and to account for violated expectations, anomalies, and "nonsensical" outcomes. All the participants found at least an image or a heatmap that did not fit with their assumptions, preliminary understandings, or ongoing theorizations (e.g., an image that could not be logically connected with its prompt). In all these cases, participants could not ignore the "case" and had to engage in a meaning-making process: they could either avoid redefining the ongoing theory, as in the confirmatory additive process, rather preferring to "make sense" of the anomaly, squeezing it

into the theoretical framework under development; or they could revise the theory, as it happened in the circular iterative process, even abandoning it and formulating ex novo a new conceptualization.

We did not force participants to engage in theory building. Nonetheless, they kept seeking for explanations throughout the whole study. This may signal how finding meaning in the behavior of the text-to-image model really mattered to them. In the context of self-tracking technologies, it has been observed that when users do not find a given technological output relevant, like certain data visualized by an activity tracker, they prefer to abandon the technology if such data are unexplainable, obscure, or apparently meaningless, rather than engaging in meaning-making (Lazar et al., 2015; Rapp & Cena, 2016; Rapp & Tirassa, 2017; Rapp et al., 2019; Attig & Franke, 2020; Rapp, 2023; Rapp & Boldi, 2023). Conversely, when the outputs matter to them, as in case of athletes that use quantified information of their sports activity to improve their performance and tune their trainings (Rapp & Tirabeni, 2018; Rapp & Tirabeni, 2020), anomalies, unclear insights, and obscure information might encourage them to make more sense.

In the context of text-to-image GenAI, the need to explain every single produced image, theorize about its possible causes, and create conceptualizations about its creator, may come from the relevance that this kind of technology may have for the participants. Even the participants who did not have any previous knowledge of and experience with GenAI strived for accounting for its outputs and its working principles. This situation reminds certain novels of science fiction writer Stanislaw Lem, like Solaris or Planet Eden, where the main characters continuously struggle to find an explanation for the behavior of alien forms of life that seemingly exhibit human-like behavior, which, nevertheless, remain essentially unknowable. It is not difficult to hypothesize that this need of sense-making stems from the peculiarity of the technology and its potential disturbing nature. A technology that performs tasks typically associated with humans or generates human-like outputs, thereby naturally evoking attributions of human-like abilities, goals, and values, which, nonetheless, on a closer look, cannot be explained in human terms, represents an enigma that cannot remain unsolved.

Moreover, text-to-image technology can create in a more efficient and productive manners than humans: this superhuman skills, that transcend ordinary human capacity (Li et al., 2014), may be perceived as threatening, requiring "conceptual tools", like the folk theories, that are able to partially soothe their unsettling effects. In fact, some of the developed theories, like those of GenAI as human-like or as human byproduct, were able to tame the participants' concerns about the nature of the technology. Others, like that of GenAI as

other-than-human, were rather associated with eerie feelings, also evoking mistrust toward the technology. A technology of this kind is perceived as threatening because may have "unknown" abilities, goals and values, misaligned from those of human beings, whereby its autonomy and independence may even put at risk human existence.

In this context, XAI tools, like the heatmaps, might calm the anguish of the unknown as well, somehow clarifying the opaque nature of the technology. We observed that the heatmaps helped participants create a general sense of the technology. However, rather than yielding an illusionary explanatory depth and an overconfidence bias, as noticed in some previous research (Rozenblit & Keil, 2002; Chromik et al., 2021), we observed that these XAI tools generated doubts or were a stimulus to revise previous theories. In fact, heatmaps revealed little about why the text-to-image model generated a specific image from a specific term in that way (Kim et al., 2023; Lee et al., 2023; Tang et al., 2023), and were limited in driving the individuals' sense-making process towards a certain direction. Therefore, if the goal is to obtain an accurate explanation geared towards a genuine understanding of how technology operates, the necessity for more advanced tools appears evident. Instead, heatmaps could be effectively used to dismantle previous theories, disconfirm assumptions, and, ultimately, dehumanize the model, showing that it does not follow our ways of perceiving and reasoning. However, they might even foster the development of "erroneous" theories about how the model came to a decision or an output, or have the opposite effect (e.g., increasing the humanization of the model): a visual tool is subject to multiple interpretations and such interpretations are currently not controlled by design.

However, we may ask ourselves why it should be important that the folk theories people develop align with the actual functioning of the technology. There are many cases in which individuals have "erroneous" theories about how a given technology works, yet this does not have any negative impact on the interaction, or, more generally, on their life (consider, for example, erroneous folk theories related to a television). In the context of GenAI, instead, and, more in general, of AI, having a "correct" theory of the inner working of the technology may be crucial. For instance, if we consider the field of collaborative decision-making, where humans and AIs may "collaborate" to make a decision (Lai et al., 2023), the folk theories that humans develop regarding the operations of the AI become fundamental. For example, if an incorrect theory is built about how an image generation model works and then the images it produces are used to make important decisions, a problem may

arise. In fact, how can we trust an "agent" and use its outputs or suggestions to make informed decisions if we cannot understand how it actually came to those outputs and suggestions?

This raises the need to consider how possible erroneous theories may be harmful for the users, as well as to increase the comprehensibility of these technologies mitigating users' biases and concerns. In the next section, we will proceed to outline a series of design suggestions based on the study findings.

6 IMPLICATIONS FOR DESIGN

Although the primary purpose of our study was not to generate design recommendations but to provide an empirical understanding of how people conceptualize a text-to-image model, we may identify two implications coming from our findings for designing novel XAI instruments.

6.1 Designing Gen-XAI adaptive tools co-evolving with users' folk theories

Previous scholarly investigations into folk theories have highlighted their often inconsistent and malleable nature (Gelman & Legare, 2011; Rader & Gray, 2015; Eslami et al., 2015; DeVito et al., 2017). Our study findings point out even more the flexibility of folk theories of text-to-image GenAI, which might be due to the specific nature of the technology (e.g., DeVito et al., 2018; DeVito, 2021): participants developed a variety of local explanations and general theories, which were even contrasting or contradictory, and tried to accommodate "new cases" by either squeezing them into a pre-existing theory, or by revising and substituting the ongoing conceptualization. In so doing, they often relied on previous knowledge, or framed the "unfamiliar technology" within more familiar frames, like supposedly similar technologies, human beings, or their own experiences and practices.

Given the extreme dynamic nature of the theory formation process, current "static" XAI methods, might be insufficient. While XAI techniques like heatmaps may provide cues that stimulate the theory formation process and increase the knowledge that people have about the technology, such knowledge may need to be integrated into their preexisting understandings, as well as evolve as new cases and difficult-to-explain outputs are encountered. This emphasizes the importance of developing XAI systems capable of adapting to and changing in synchrony with the user's evolving theorizing over time, also taking into account their preexisting knowledge and frames through which they explain the technology. In this sense, designing "adaptive XAI" tools (Robrecht et al., 2023) appear fundamental. These tools might integrate tailored explanations on the basis

of users' specific needs and characteristics (Kim et al., 2023; Lee et al., 2023; Ehsan & Riedl, 2020) with real-time adaptations (Robrecht et al., 2023; Moore et al., 2023).

In this sense, our study findings emphasize that a one-size-fits-all approach may be not beneficial, as people with different backgrounds formulate different interpretations of text-to-image models and may need diverse explanations to understand their functioning. However, although first approaches to this have been presented (Robrecht & Kopp, 2023), implementing an adaptive explainer remains a challenge.

Moreover, our study findings highlight that the explanations should be adapted not only to the user's characteristics and needs, but also to their pre-existing and evolving knowledge and theories through which they attempt to explain the technology.

In both cases, recent advances in Large Language Models (LLMs) may facilitate the adaptation endeavor (Mavrepis et al., 2024). LLMs could be primarily used to encourage users to express their preexisting assumptions and understandings, as well as those that may emerge in the ongoing interaction. Then, by capturing such conceptualizations, XAI systems could adapt their explanations in terms of style and content. For instance, they could incorporate metaphors, analogies, and examples that resonate with users' existing concepts and previous experiences; or remain receptive to new folk theories that may stem from system use over time. In other words, adaptive XAI methods could gradually modify their explanations based on information continuously expressed by the user, allowing for a gradual co-construction of meaning.

Nonetheless, there is a need to strike a delicate balance. On the one hand, individuals tend to rely on familiar conceptual frameworks to interpret and explain technology (e.g., by comparing the work of the GenAI to the designer's work), which is also confirmed by previous research (French & Hancock, 2017). This suggests that it might be useful to use familiar metaphors that evoke already known processes to facilitate comprehension of complex technology. On the other hand, there is a risk that this could promote the formulation of "erroneous" theories. In our study, for example, we have noticed that participants may tend to generate "erroneous" folk theories of text-to-image models (e.g., believing that they are a sort of search engine that retrieves pre-existing images from a repository), which, in certain cases may be counterproductive: for example, people may not be able to obtain from the technology what they want, if they believe that it works in a way that does not correspond to its real functioning. To address this issue, it could be advantageous to gradually guide individuals

in understanding aspects of the system, starting from their preexisting frames, knowledge and assumptions, and progressively separate erroneous concepts from more "genuine" interpretations.

6.2 Designing XAI tools acknowledging GenAI intrinsic machine-nature complexity

Research in XAI aims, by definition, to make the inner working of AI technologies easier to be interpreted (Ehsan & Riedl, 2020; Speith, 2022; Capel & Brereton, 2023; Hoffman et al., 2023). As previously noted, the extreme complexity of their functioning and their inherently opaque nature pose significant challenges, with some aspects inevitably remaining obscure (Tang et al., 2023; Lee et al., 2023; Alqaraawi et al., 2020). However, providing tools that ease the understanding of these systems could ironically increase the illusion of explanatory depth (Rozenblit & Keil, 2002; Chromik et al., 2021).

Our study findings highlight that participants tried to trace the unfamiliar of the inexplainable back to the familiarity of known technologies or "beings", like other humans or themselves. In so doing, however, they often oversimplified the functioning of the text-to-image model or produced misleading explanations. In particular, participants engaged in either anthropomorphizing the GenAI (GenAI as human-like, or GenAI as human byproduct), or in considering it as an alien entity (GenAI as other-than-human). On the one side, the former conceptualizations may be misleading, making the individuals believe that the text-to-image model has "cognitive" and "perceptual" abilities that are similar to ours. This may produce harmful effects in certain contexts, for example in co-decision making and cooperation: believing that our "partner" fundamentally follows our same rational processes, while it actually functions in a completely different way, may lead to incorrect assumptions about why it has suggested/done what it has suggested/done, resulting in erroneous decisions. On the other hand, seeing the GenAI as an alien, thus extraneous, yet intelligent and intentional, may provoke unsettling emotions and mistrust, which also may undermine collaboration and co-decision.

In this scenario, it is crucial to convey that "complex" does not necessarily equate to "alien". Indeed, the complexity might simply differ from our own, representing a level of sophistication that we, as humans, may not fully comprehend in detail but can, nonetheless, recognize as complex. In this sense, designers could experiment with XAI tools that intentionally avoid offering exhaustive explanations of system functioning. Instead, they may strategically introduce defamiliarization strategies to counterbalance oversimplistic conceptualizations and reveal contradictions in misleading folk theories, thus acknowledging the inherent complexity of the system which may encapsulate blind spots. This approach might be particularly beneficial

when users begin to develop an illusion of explanatory depth or exhibit tendencies to anthropomorphize the GenAI, which may lead them to overlook its machine nature.

The study findings showed that the heatmaps we employed could be able to challenge or disconfirm pre-existing theories regarding the text-to-image model's functioning. From this perspective, they could even be used as tools for "dehumanizing the technology", discouraging the tendency that we found across all the participants to consider the model endowed with abilities that are similar to ours. However, not being intentionally designed to disconfirm pre-existing beliefs, heatmaps may also be used for the opposite aim, thus confirming assumptions, or even increasing the humanization of the technology.

Future research lines could experiment with novel XAI techniques that incorporate ambiguity in their designs, possibly balanced with a certain degree of "control" on the interpretations produced by users. Ambiguity has been seen as an insightful design resource, stimulating open-ended, multiple, and unexpected interpretations (Gaver et al., 2003). XAI tools intentionally designed to be ambiguous could better encourage individuals to question their assumptions and previous theorizations. These could be delivered when the LLM detects that the user has developed a humanizing theory of the technology. The ambiguous representations of the text-to-image GenAI's behavior could also be used as a projective method to favor the users' expression of their interpretations, which can be captured by the system, and then used to formulate tailored explanations. This process could be repeated periodically, whereby the system could present intentionally counterintuitive "explanations" or representations of its inner behavior to make the user perceive its inherent complexity (Rapp et al., 2025a; Di Lodovico et al., 2025). The LLM could then check the folk theory that the ambiguous representation has contributed to elicited, to ensure that it is guiding the user's theorization in the right direction. Otherwise, another ambiguous representation could be delivered.

7 LIMITATIONS

Our study focused on a specific GenAI text-to-image model, potentially limiting the generalizability of the findings. However, the specific model was not revealed to the participants, who generated their theories starting only from its outputs. This may increase the generalizability of the findings, with reference to the theories generated, which could be applicable to other GenAI text-to-image models. Additionally, it is debatable whether the study conclusions can be applied to diverse cultures, given the study's confinement to the Italian population. The absence of direct interaction between participants and GenAI prevented us from

exploring how participants developed folk theories while generating images autonomously by using the model. However, people tend to formulate their own theories about phenomena even without directly experiencing them (Chromik et al., 2021). It is reasonable to assume that the findings from our study could extend to self-created images, as participants with prior experience in text-to-image models developed theories similar to those without such experience. Future research could deepen the study of folk theories about GenAI, comparing the conceptualizations developed during the interaction with those presented in this article: this could lead to discover "interaction strategies" that participants could use to discover the functioning of the model (e.g., experimenting different prompts as stimuli for understanding how the technology works) and that we could not observe in this study. Finally, participants might have formulated theories regarding the operation of GenAI, being shaped by the context and materials used in our study. By asking them to determine the subject of each image, their focus could be possibly directed towards a specific part of the generation process, that is the transformation of the prompt into the intended subject. However, all the participants went beyond this part of the generation process and developed a variety of theories that also conceptualized the GenAI as an intentional entity. Moreover, the choice of different images (and terms that originated them) could have potentially elicited different theories. However, given the large number of images provided, their heterogeneity and their comprehensibility, we believe that we covered a substantial range of possible cases.

8 CONCLUSION

In this article, we explored what kind of folk theories individuals may develop in relation to GenAI text-to-image models, as well as how these conceptualizations are formed.

We discovered that participants built both mechanistic theories focusing on the GenAI's functional operations, and essentialist theories, attributing mental states, like intentionality and agency, to the technology. Moreover, we showed that folk theory development unfolds mainly through trial and error, whereby participants followed either a linear additive process, in which each step of a theory is grounded on the previous one, or a circular iterative process, where phases of theory building are alternated with moments of assessment, refinement, or substitution. Finally, we pointed out the diverse strategies that participants used to explain the GenAI's functioning, which ultimately attempt to trace the unexplainable nature of the technology back to more familiar frames.

These findings entail three main contributions. First, we identify folk theories that were not pointed out by previous research, like GenAI as a human byproduct and GenAI as other-than-human, which appear peculiar to GenAI technology. Second, we detail how people generate folk theories about technology, providing insights about the theory building process that were not highlighted by previous research, like the extreme malleability of both the theory formation process and the theories produced. Third, we emphasize that people with different backgrounds may have different attitudes towards theory development, whereby GenAI peculiarities may foster a generalized tendency to anthropomorphize its behavior.

CONTRIBUTION STATEMENT

Chiara Di Lodovico: Investigation, Conceptualization, Methodology, Formal analysis, Data Curation, Writing – original draft; **Federico Torrielli**: Software; **Luigi Di Caro**: Resources, Methodology, Writing – review & editing; **Amon Rapp**: Conceptualization, Methodology, Formal analysis, Validation, Writing – original draft, Writing – review & editing, Project administration, Supervision.

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